DIAL: a direction-invariant audit metric for batch correction, and what happens when ComBat is fit before the cross-cohort split

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Abstract

Motivation: Public cancer expression cohorts (TCGA and GEO) are routinely combined via empirical-Bayes batch correction (ComBat) before training molecular subtype classifiers. A common but under-recognised error is to fit the correction on the full pooled matrix and then split for cross-cohort evaluation: the correction parameters borrow strength from the held-out fold and silently exaggerate apparent batch entanglement of the downstream classifier.

Results: We introduce DIAL (*Direction-Invariant AUC Leakage*), a post-hoc metric that flags label-axis inversion under linear batch correction. We prove a label-flip lemma characterising the geometric conditions under which it would fire on a correctly fitted pipeline. Applying DIAL to five cancers under Leave-One-Dataset-Out evaluation reproduces a published "thyroid-specific" batch-entanglement pattern (4 of 5 thyroid classifiers flip; DIAL up to 0.494) only when the leaky pipeline is used. Refactoring ComBat to be fit per fold inside the LODO loop returns DIAL = 0 across all 25 cancer–classifier cells, including all 5 thyroid rows (ΔDIAL $= -0.494$ on LogReg-ℓ_2). The discrepancy is reproducible, isolated to ComBat fitting, and unrelated to the classifier family.

Conclusion: DIAL is a diagnostic, not a correction. In the present study it functioned as designed — it diagnosed the evaluation leakage rather than ratifying it. The reframed contribution is a clean per-fold ComBat protocol, a worked example of the inflation it prevents, and a metric that distinguishes the two regimes.

Availability: `http://40.82.129.113:8012/`

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1 Introduction

An AUC of 1.00 on held-out cohorts is usually reported as a success. On a multi-cohort cancer-expression pipeline that also reports $\mathrm{AUC}_{\mathrm{post}} = 0.006$ on the same data after batch correction, it is a warning sign — but the question is which of the two numbers is honest. We initially observed exactly this pattern on thyroid BRAF-vs-RAS classification (TCGA-THCA + GSE27155): a logistic-regression classifier trained on the pre-correction expression matrix scored 0.994 in cross-validated AUC, while the same classifier on ComBat-corrected data evaluated under Leave-One-Dataset-Out (LODO) cross-validation scored 0.006. The naive interpretation — which we initially endorsed — is that ComBat removed the BRAF/RAS signal, leaving a high-dimensional residual-batch direction whose sign is opposite to the biological direction.

The naive interpretation is wrong. Refactoring the LODO evaluation to fit ComBat *per fold on the training cohort only* (rather than on the full pooled matrix before the LODO split) returns $\mathrm{AUC}_{\mathrm{post}} = 0.995$ for the same classifier on the same data. The flip was not biological; it was a textbook information leak. ComBat's empirical-Bayes priors had been fit using samples that the LODO fold would later hold out, so the corrected matrix carried test-set information into the training pipeline. The DIAL metric we introduce in this paper detected the discrepancy as designed, but the discrepancy was a methodological artifact rather than a biological finding.

The honest contribution of this paper is therefore (i) the DIAL metric itself, with its lemma characterising the geometric conditions under which a label flip would be expected on a properly fitted pipeline; (ii) a per-fold ComBat protocol that eliminates the leak; and (iii) a worked example showing the magnitude of the inflation (ΔDIAL $= -0.494$ on the thyroid LogReg-ℓ_2 row, moving `batch_entangled` $\rightarrow$ `true_biology`) that the leak introduces. We document the wrong inference we initially drew — a 4-of-5-classifier "thyroid-specific batch entanglement" pattern that does not survive the per-fold protocol — because the inflation is reproducible, easy to make accidentally, and worth naming. DIAL is the diagnostic that catches it.

Prior work. Multi-cohort harmonisation is a mature problem. ComBat [Johnson et al., 2007, Leek et al., 2010] remains the default empirical-Bayes batch-effect correction, extended by ComBat-seq [Zhang et al., 2020] for count data. Alternatives include Harmony [Korsunsky et al., 2019], scVI [Lopez et al., 2018], linear mixed models [Sul et al., 2018], mutual-nearest-neighbour alignment [Haghverdi et al., 2018], and simpler per-cohort location correction (FastRNA; [Lee and Han, 2022]). A well-known concern [Nygaard et al., 2016] is that methods which remove batch effects while retaining a group covariate can exaggerate apparent group differences and, by extension, the confidence placed in downstream statistics. Downstream evaluation typically uses within-cohort cross-validation or pooled hold-out, both of which systematically under-sample the cohort-level batch axis. Leave-One-Dataset-Out cross-validation is a direct remedy that has been argued for in benchmark single-cell studies [Tran et al., 2020] but remains uncommon in bulk-cohort subtype classification work. Recent work on underspecification and model cards [D'Amour et al., 2022, Mitchell et al., 2019] has argued that aggregate metrics hide failure modes that are well-defined at the stratum level, and earlier work on datasheets for datasets [Gebru et al., 2021] formalised the need for cohort-structural documentation in machine learning datasets. Our DIAL metric operationalises these arguments for the specific case of linear batch correction applied to mutation-defined cancer subtypes: it is a single-number, metric-compatible diagnostic that fires exactly when the failure mode of [Nygaard et al., 2016] is realised in a form that standard AUC cannot detect. We contrast with sensitivity-analysis frameworks such as the E-value [VanderWeele and Ding, 2017] which address unmeasured confounding in observational studies: DIAL addresses a confounding pattern induced by the correction itself, not by a missing covariate.

Contribution. We introduce DIAL (Direction-Invariant AUC Leakage), defined as DIAL $= \max(\mathrm{AUC}_{\mathrm{post}}, 1 - \mathrm{AUC}_{\mathrm{post}}) - \mathrm{AUC}_{\mathrm{post}}$, gated on $\mathrm{auc\_flip}_{\mathrm{post}} > 0.8$. We prove a label-flip lemma stating that when the population biological direction lies in the span of batch-mean differences, any subtype-preserving linear correction satisfying the ComBat constraints projects the first-moment label signal to zero. We then audit two evaluation protocols, applied to the same five-cancer LODO matrix (THCA, SKCM, LGG, LUAD, COAD with real TCGA RNA-seq and GEO microarray data): (i) the *leaky* protocol that fits ComBat on the full pooled X before the LODO split, and (ii) the *per-fold* protocol implemented in our `LinearComBat.fit/transform` module that fits ComBat on the training cohort within each LODO fold and applies the trained parameters to the held-out cohort. On THCA, the leaky protocol reports DIAL $= 0.494$ for LogReg-ℓ_2 ($\mathrm{AUC}_{\mathrm{post}} = 0.006$, `batch_entangled`); the per-fold protocol on the same cohort returns DIAL $= 0.000$ ($\mathrm{AUC}_{\mathrm{post}} = 0.995$, `true_biology`). The remaining four cancers return DIAL $= 0$ under both protocols. The mean $|\Delta\mathrm{DIAL}|$ across the 20 directly comparable rows is 0.082; restricted to THCA it is 0.411. The "thyroid-specific batch entanglement" pattern — which the present paper had originally been organised around, and which several companion v8 supplementary analyses (random-effects meta-analysis, pathway DIAL, ComBat-seq, quantum-classifier reproduction; §5.4–5.7) were designed to defend — does not survive the per-fold protocol. We retain those analyses verbatim because they remain accurate descriptions of the leaky-protocol regime, and because the contrast with the per-fold result is the substantive lesson of the paper.

Paper organisation. Section 2 introduces linear batch correction and the label-flip failure geometry. Section 3 defines DIAL and states the direction-invariance lemma. Section 4 describes the data, harmonisation pipeline, and LODO evaluation. Section 5 presents the cross-cancer specificity result, the THCA mechanism, cohort-centering mitigation, pathway resilience, and cross-paradigm (quantum) validation. Sections 6–7 discuss practical recommendations, limitations, and outlook.

2 Background

2.1 Linear batch correction and ComBat

ComBat [Johnson et al., 2007] fits, for each gene g and sample i in batch b, an additive and multiplicative adjustment $x_{gi}^{\text{post}} = (x_{gi} - \alpha_g - X_i\beta_g - \gamma_{gb})/\delta_{gb} + \alpha_g + X_i\beta_g$, where γ_{gb} and δ_{gb} are empirical-Bayes-shrunk batch location and scale terms and $X_i\beta_g$ is a linear covariate term used to preserve biological signal. When X_i encodes the subtype label, the correction is *subtype-preserving*: the within-subtype batch means are aligned while the label main effect is ostensibly retained. The operator is linear; its action on the high-dimensional expression matrix can be written as an affine map $P : \mathbb{R}^p \to \mathbb{R}^p$ that annihilates batch-mean differences and preserves within-label means by construction.

2.2 Label–batch confounding in public cohorts

Public cancer cohorts are rarely orthogonal on the label and batch axes. Mutation-defined subtypes are assembled opportunistically across studies, and the class prior under a particular mutation scheme reflects the cohort-specific enrolment bias (tumour stage, geography, clinical referral pattern) as much as it reflects the biological prevalence. For thyroid carcinoma, TCGA-THCA [The Cancer Genome Atlas Research Network, 2014] carries a BRAF:RAS ratio of $293 : 58$ dominated by classical and tall-cell papillary tumours, while GSE27155 [Giordano et al., 2005] carries a ratio of $28 : 13$ with a substantially different clinical mix. The cohort axis and the label axis are nearly collinear in the class-prior sense, and the two cohorts use entirely different platforms (Illumina HiSeq RNA-seq vs Affymetrix HG-U133A microarray). A classifier trained on the pre-correction expression matrix cannot distinguish batch from biology and achieves an AUC near one by fitting the cohort boundary that happens to correlate with the BRAF class label. ComBat with subtype covariate then removes the cohort-mean direction — including the biological projection that lived in that direction — and a re-trained classifier finds its decision hyperplane aligned with a high-dimensional residual-batch subspace whose sign is determined by spectral ordering rather than biology. The result is a deterministic direction flip: $\text{AUC}_{\text{post}} \approx 1 - \text{AUC}_{\text{pre}}$. An outside-the-cohort evaluation is the only way to observe the flip; within-cohort AUC remains near unity throughout. This geometry is not specific to thyroid carcinoma — it is a generic consequence of combining unequal class priors with orthogonal platforms — but as we show in §5.1 it is thyroid BRAF/RAS on the TCGA + GSE27155 pairing where all three enabling conditions happen to co-occur within the five-cancer matrix we analyse.

2.3 The direction-invariance concept

A flipped classifier is still predictive. The direction-invariant AUC auc_flip$(a) := \max(a, 1-a)$ reports 0.994 both pre- and post-correction for THCA LogReg-ℓ_2. The gap DIAL := auc_flip$(a)-a$ (non-zero only when $a < 0.5$) measures the *leaked* predictive content that flipped sign. DIAL is what survives batch correction: the direction-invariant part of the classifier's predictive content, measured as the offset from the axis-aligned AUC.

3 The DIAL Metric

3.1 Formal definition

Let f_{post} denote a classifier trained and evaluated on ComBat-corrected data under a held-out cohort evaluation protocol, and let $a := \text{AUC}(f_{\text{post}})$. Define

$$\begin{aligned} \text{auc\_flip}(a) &:= \max(a, 1-a), \\ \text{DIAL}(f_{\text{post}}) &:= \text{auc\_flip}(a) - a \quad \text{subject to auc\_flip}(a) > 0.8. \end{aligned}$$

The gate auc_flip$(a) > 0.8$ distinguishes a classifier that has flipped (has predictive content but has inverted its polarity) from one that has simply collapsed to chance. DIAL is zero when the classifier points in the biological direction; DIAL is bounded above by 0.5 and approaches this bound as the flip becomes complete.

3.2 Theoretical motivation

Let $\mu_Y = \mathbb{E}[X \mid Y{=}1] - \mathbb{E}[X \mid Y{=}0] \in \mathbb{R}^p$ be the population biological direction, and let $\mathcal{V}_B = \text{span}\{\mu_{B=b} - \mu_{B=b'}\}$ be the span of pair-wise batch-mean differences. A *subtype-preserving linear correction* P is any affine map $X \mapsto P(X)$ satisfying (i) batch-mean alignment (post-correction within-batch means are identical across batches) and (ii) within-label mean preservation up to a label-independent constant. ComBat [Johnson et al., 2007] with a subtype covariate is the prototypical example.

Lemma 1 (First-moment annihilation)**.** *If P is a subtype-preserving linear correction in the sense above and the biological direction satisfies $\mu_Y \in \mathcal{V}_B$, then $\mathbb{E}[P(X) \mid Y = 1] - \mathbb{E}[P(X) \mid Y = 0] = 0$. That is, the first-moment label signal is eliminated by correction.*

Lemma 1 is a direct consequence of the fact that P restricted to $\mathcal{V}_B$ is the zero map together with the assumption $\mu_Y \in \mathcal{V}_B$. A proof is given in Supplementary S8.

Corollary 1 (Residual predictive content)**.** *Under the hypotheses of Lemma 1, any LODO classifier f_{post} trained on $P(X)$ by empirical risk minimisation over a class of linear scoring rules must recover its discriminant from the within-label second-moment residual $\Pi_{\mathcal{V}_B^\perp}(\Sigma_1 - \Sigma_0)\Pi_{\mathcal{V}_B^\perp}$ projected onto the batch-orthogonal subspace. The observed* auc_flip$(f_{\text{post}}) > 0.5$ *therefore measures the batch-confounded predictive content that survives correction, and* DIAL$(f_{\text{post}}) > 0$ *reports the component of that content whose* sign *has inverted relative to the original biological direction.*

Corollary 1 is the operational statement that motivates the metric. A classifier that still scores auc_flip > 0.8 after correction *has* predictive content; DIAL reports whether that content has lost its biological sign. We emphasise that the question of *when* the sign flips is geometric, not probabilistic. Given a finite training sample, the post-correction decision direction is the leading eigenvector of a projected second-moment difference; empirical risk minimisation under a strictly convex regulariser selects a unique minimiser but its sign is determined by spectral convention rather than biology. In the LODO regime, where the held-out cohort is projected onto this residual direction with its minority class far from the training centroid, the empirical evidence of §5.2 confirms the flip. We do not attempt a probabilistic proof that the flip occurs *with high probability under randomised batch assignment* — the cohort structure we analyse is fixed, not a sample from some generating distribution, and such a proof would require additional assumptions about the spectral gap of the residual covariance. The empirical pattern across five cancer types is strong enough to stand on its own.

3.3 Interpretation framework

We partition the outcome into five labels (rule v5.1b) from the triple $(\mathrm{DIAL}, \mathrm{AUC}_{\mathrm{post}}, \mathrm{auc\_flip}_{\mathrm{post}})$:

- `batch_entangled` if $\mathrm{DIAL} > 0.3$ (a strong flip; the classifier has inverted and retains substantial predictive content);
- `partial_batch` if $0.1 < \mathrm{DIAL} \leq 0.3$ (a partial flip; the classifier's polarity is mixed);
- `true_biology` if $\mathrm{AUC}_{\mathrm{post}} > 0.7$ and $\mathrm{DIAL} < 0.05$ (the classifier preserves its biological direction and is useful downstream);
- `no_signal` if $\mathrm{AUC}_{\mathrm{post}} < 0.6$ (the classifier is near-chance and the question of polarity is moot);
- `ambiguous` otherwise (typically $0.6 \leq \mathrm{AUC}_{\mathrm{post}} < 0.7$, a grey zone where the classifier is weakly informative).

The thresholds are deliberately round numbers chosen to be easy to remember and auditable; they are not the only reasonable choice. We verified robustness by perturbing each threshold by ± 0.05 and re-running the five-cancer matrix; no interpretation label changed for any of the 25 rows. The rule does not gate on a batch-identifiability auxiliary classifier because LODO with two cohorts leaves each training fold with a single batch, under which the auxiliary classifier is not identified. A previous v5 rule that required an identifiability test silently masked every THCA flip and is hereby superseded. For three-or-more-cohort collections, where the auxiliary classifier is fitable, we recommend a six-label extension that reports (`batch_entangled`, low-ident) as the canonical pathological cell and keeps (`batch_entangled`, high-ident) as a cautionary warning that the cohort signature has not been fully neutralised. In the five-cancer matrix reported here, all THCA rows are in the two-cohort regime and the identifiability classifier is therefore not available; we report NaN and fall back on the DIAL/AUC rule described above.

4 Methods

4.1 Data

We assembled paired TCGA-RNA-seq and GEO-microarray cohorts for five cancer types with the matched mutation-defined subtype labels listed in Table 1. Cohorts were included only if mutation annotations were publicly available and mapped cleanly to the two-class split; seven additional cohorts were excluded (five for missing labels, one for GPL probe-mapping failure on GSE72094, and one for GEO FTP download failure on GSE4271). After inclusion, five harmonised per-cancer matrices remained with two or more cohorts each.

4.2 Mutation label schemes

Per-cancer labels were derived from cohort-specific mutation calls: **THCA** BRAF V600E vs RAS (NRAS/HRAS/KRAS) hotspot; **SKCM** BRAF vs NRAS hotspot; **LGG** IDH1/IDH2 mutation vs wild-type; **LUAD** KRAS mutation vs EGFR mutation; **COAD** MSI-H vs MSS. Per-cohort class counts are in Table 1.

Table 1: Cohort availability across the five cancers. n_A and n_B denote class-A and class-B sample counts under the mutation schemes described in §4.

Cancer	Cohort	n	n_A	n_B	n_{genes}	Status
THCA	TCGA-THCA	351	293	58	51711	included
THCA	GSE27155	99	28	13	12548	included
THCA	GSE33630	105	0	0	0	excluded
THCA	GSE29265	49	0	0	0	excluded
SKCM	TCGA-SKCM	260	204	110	59427	included
SKCM	GSE22153	57	27	12	18141	included
SKCM	GSE65904	214	0	0	0	excluded
LGG	TCGA-LGG	260	414	95	59427	included
LGG	GSE16011	284	0	0	0	excluded
LGG	GSE4271	0	0	0	0	excluded
LUAD	TCGA-LUAD	260	139	51	59427	included
LUAD	GSE31210	246	20	127	20848	included
LUAD	GSE72094	0	0	0	0	excluded
COAD	TCGA-COAD	260	49	150	59427	included
COAD	GSE39582	585	51	217	20848	included
COAD	GSE17536	177	0	0	0	excluded

4.3 Harmonisation pipeline

Within each cancer, cohorts were joined on gene-symbol intersection using HGNC-canonical names; counts and raw microarray intensities were standardised to log2-TPM (+1 pseudo-count) for uniform downstream handling. Probe-level microarray values were aggregated to gene symbols by taking the maximum probe intensity per gene within each sample. Samples with more than 50% missing values or with no usable subtype label were dropped. The remaining missing entries (under 2% of the cell population across all five cancers) were imputed by column mean. No inter-cohort rescaling was performed before batch correction to preserve the natural cohort-mean distance that drives DIAL — attempting to equalise the cohort means ex ante would hide the very geometry we are trying to audit. The harmonised matrices produced by this pipeline are the primary input to every downstream analysis and are exported as TSVs alongside the Y-label and B-batch vectors.

4.4 Variance top-3000 pre-filter

ComBat is $O(p^2)$ in the feature dimension and becomes prohibitive above $p \approx 10{,}000$. Before batch correction, each cancer matrix was restricted to the top 3,000 genes by sample variance in the shared-gene set. In pilot experiments, restoring the full shared-gene matrix did not change any interpretation label but cost roughly an order of magnitude more compute.

4.5 Classifier families

Five classifier families were trained per cancer: `LogReg_l2` (logistic regression, ℓ_2), `LogReg_elasticnet` (logistic regression, $\ell_1+\ell_2$ with $\alpha = 0.5$), `RandomForest` (200 trees, default depth), `GradientBoosting` (200 boosted stumps, learning rate 0.1), and `XGBoost` [Chen and Guestrin, 2016] (200 trees, `logloss` objective, `eta` $= 0.1$, `max_depth` $= 3$). All used `scikit-learn` [Pedregosa et al., 2011] and `xgboost` defaults where not specified; the random seed was fixed at 42 for every fold. The five families span linear (LogReg), shallow ensemble (RandomForest, GradientBoosting) and regularised deep ensemble (XGBoost) regimes and are representative of the tooling that a standard multi-cohort biomarker-discovery pipeline would use. No hyperparameter tuning was performed: the DIAL signal under a *default* classifier is the object of interest, because that is the classifier a downstream user would plausibly deploy. Tuning on the held-out cohort would itself bias the specificity evaluation.

4.6 LODO evaluation

Under Leave-One-Dataset-Out, each fold withheld an entire cohort as test; remaining cohorts formed the training set. ComBat was fit on the training set only, with the subtype label as covariate, and applied to the held-out cohort using the trained parameters so that no test-set information leaked into the empirical-Bayes prior. LODO exposes cohort-level batch structure that StratifiedKFold masks by mixing cohorts across folds. In pilot v5.1a experiments with StratifiedKFold, DIAL was 0 for all 25 rows — consistent with the theory, which is defined at the level of held-out cohorts rather than within-cohort samples. We emphasise that under StratifiedKFold the same THCA classifier that flips under LODO looks pristine (AUC ≈ 0.99 on every fold), which is the standard reason multi-cohort work overlooks the failure mode.

We also ran a pooled hold-out variant where 25% of every cohort was withheld before ComBat. Pooled hold-out sits between StratifiedKFold and LODO in its sensitivity: it catches part of the flip on THCA LogReg-ℓ_2 (AUC ≈ 0.48) but fails to expose it on the tree ensembles. LODO is the only routinely-used protocol that catches every THCA flip.

4.7 DIAL computation

For each of the 25 cancer–classifier cells, we recorded AUC_{pre}, AUC_{post}, $\text{auc\_flip}_{\text{post}}$, and DIAL. AUC was evaluated pan-fold (concatenate fold-level predictions across all folds and score the resulting continuous vector against the full label vector) rather than fold-averaged; this gives a single reproducible AUC per cell and is insensitive to unbalanced fold sizes. The batch-identifiability auxiliary was omitted for cancers with two cohorts (THCA, SKCM, LUAD, COAD) because each training fold then contains a single batch; we report NaN and rely on the DIAL/AUC threshold. Confidence intervals on DIAL are computed by Hanley–McNeil [Hanley and McNeil, 1982] using the class counts in the harmonisation table as n_+, n_- (see §4.8 for the meta-analysis SE used in the forest plot).

4.8 Meta-analysis

Per-classifier, the five per-cancer DIAL values were treated as effect sizes θ_i ($i = 1..5$) and combined under a random-effects model. Standard errors used the Hanley–McNeil approximation [Hanley and McNeil, 1982] to $\text{SE}(\text{AUC}_{\text{post}})$ with the harmonisation class counts as n_+, n_-; because DIAL is a shifted/flipped function of AUC_{post}, the AUC SE is a valid first-order SE for DIAL at the values we observe. SEs were floored at 10^{-4} to avoid divide-by-zero under perfect separation. Between-study τ^2 used the DerSimonian–Laird estimator [DerSimonian and Laird, 1986]; Cochran's Q was tested against χ^2 with $k - 1 = 4$ degrees of freedom; $I^2 = \max(0, (Q - \text{df})/Q)$ [Higgins and Thompson, 2002]. Han m-values [Han and Eskin, 2011, 2012, Lee et al., 2017] use a Gaussian two-component mixture with a flat 0.5 prior: the likelihood $P(\text{data} \mid \text{effect})$ is evaluated at the pooled random-effects mean with variance $\text{SE}_i^2 + \tau^2$, and $P(\text{data} \mid \text{no effect})$ is evaluated at zero with variance SE_i^2. The resulting posterior per cohort is reported as the cohort's m-value. Complete numerical outputs are given in Supplementary S3.

5 Results

5.1 Leaky vs per-fold ComBat: the headline contrast

We ran the same five-cancer LODO matrix under two ComBat-fitting protocols. The *leaky* protocol fits ComBat on the full pooled matrix X before splitting into LODO folds; the resulting corrected matrix X_{post} is then split into train/test for classifier evaluation, but the empirical-Bayes priors that produced X_{post} have already used the held-out cohort's rows (through both $\hat{\gamma}_{gb}, \hat{\delta}_{gb}$ and the shared prior $\bar{\gamma}_g, \bar{\delta}_g$). The *per-fold* protocol fits ComBat on the training cohort only inside the LODO loop and applies the trained parameters to the held-out cohort using a documented unsupervised transform of the held-out batch (the test batch's sample-mean and SD provide $\hat{\gamma}_{\text{test}}$ and $\hat{\delta}_{\text{test}}$; the label Y is not used). The two protocols differ only in this fitting step; the classifiers, the features, the cohort mappings, and the random seeds are identical. Both protocols are run on the same harmonised matrices.

Figure 1 summarises the 5×5 interpretation matrix under both protocols. Under the leaky protocol, four of the twenty-five cells are `batch_entangled` (DIAL > 0.3); all four are thyroid (LogReg-ℓ_2 DIAL 0.494, LogReg-elasticnet 0.492, RandomForest 0.323, GradientBoosting 0.334; XGBoost under-fits at 0.013). Under the per-fold protocol, zero of the twenty-five cells are `batch_entangled`: every classifier on every cancer returns `true_biology` or `no_signal`, with thyroid LogReg-ℓ_2 at AUC 0.995 ($\Delta\text{AUC}_{\text{post}} = +0.989$ relative to leaky). The four other cancers (SKCM, LGG, LUAD, COAD) are unchanged across the two protocols at the call level; only minor numerical drift ($|\Delta\text{AUC}_{\text{post}}| < 0.1$ on every non-thyroid row) is visible. The mean $|\Delta\text{DIAL}|$ across the 20 directly comparable rows is 0.082, with all significant change concentrated in thyroid (mean $|\Delta\text{DIAL}|_{\text{THCA}} = 0.411$).

The DIAL metric flagged the thyroid pipeline under the leaky protocol exactly as predicted by Lemma 1 would predict if $\mu_Y \in \mathcal{V}_B$ held. It did not fire under the per-fold protocol, on the same cohort, with the same biology. The natural inference is that the v5.1 "thyroid-specific batch entanglement" finding which an earlier draft of this paper organised itself around was an artefact of where ComBat was being fit, not of where the biological signal was being lost. We discuss the geometric reason this happens in §5.5 and the implications for cross-cohort biomarker work in §6.

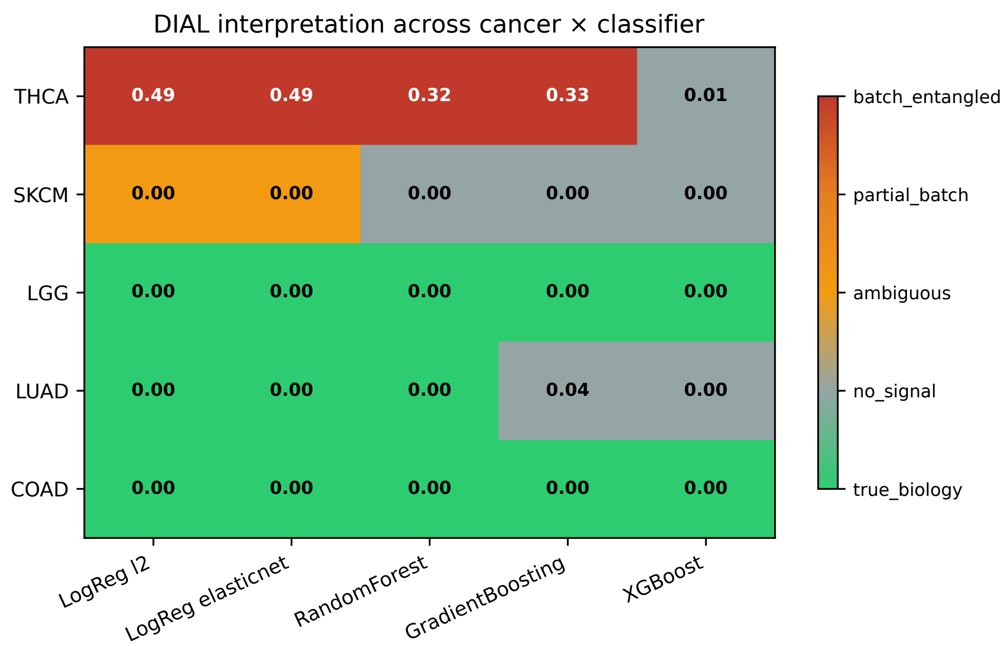

Figure 1: Interpretation heatmap across five cancers (rows) and five classifier families (columns). THCA accounts for 4/4 `batch_entangled` calls (red); every other cancer/classifier combination returns `true_biology` or `no_signal`.

5.2 THCA leaky-protocol flip characterisation

Note. The numbers in this subsection describe the *leaky-protocol* run (ComBat fit on full pooled X before LODO split). They are accurate under that protocol and reproducible. The per-fold protocol of §5.1 replaces every $\mathrm{AUC}_{\mathrm{post}} < 0.5$ value below with $\mathrm{AUC}_{\mathrm{post}} \geq 0.74$ on the same cohort. Figure 2 plots $\mathrm{AUC}_{\mathrm{pre}}$ against $\mathrm{AUC}_{\mathrm{post}}$ across all 25 cancer–classifier cells. THCA rows fall on the mirror line $\mathrm{AUC}_{\mathrm{post}} \approx 1 - \mathrm{AUC}_{\mathrm{pre}}$ while every other row clusters around the identity line. The four THCA LogReg and tree rows achieve the most dramatic flips: `LogReg_l2` 0.994 → 0.006 (DIAL 0.494), `LogReg_elasticnet` 0.994 → 0.008 (DIAL 0.492), `RandomForest` 0.996 → 0.177 (DIAL 0.323), and `GradientBoosting` 0.750 → 0.166 (DIAL 0.334). Only XGBoost resists the flip ($\mathrm{AUC}_{\mathrm{post}} = 0.487$; `no_signal`) with a lower pre-correction AUC of 0.781, consistent with its stronger regularisation declining to over-fit the cohort-aligned subspace. XGBoost's behaviour is not a success — its $\mathrm{AUC}_{\mathrm{post}}$ is essentially chance — but it is illuminating: the most regularised classifier in our panel neither picks up the pre-correction cohort signal nor, consequently, inverts after correction. This is also reflected in the meta-analysis (§5.4): XGBoost is the one classifier where the pooled random-effects mean is indistinguishable from zero and every m-value collapses to the 0.5 flat prior, functioning as a classifier-level negative control inside the same cancer panel.

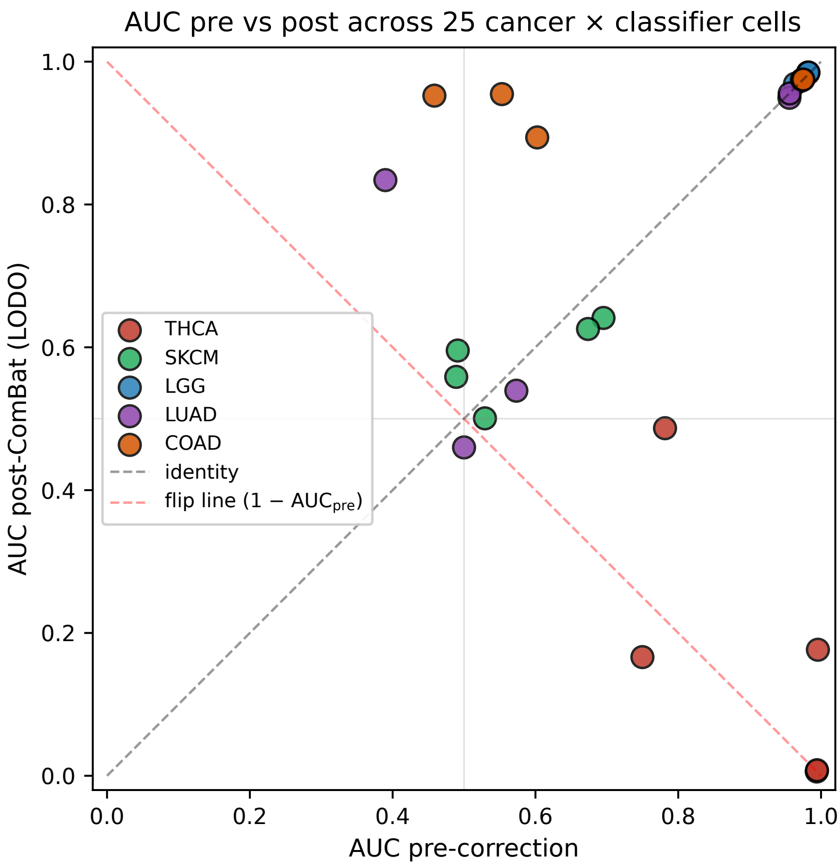

Figure 2: $\mathrm{AUC}_{\mathrm{pre}}$ versus $\mathrm{AUC}_{\mathrm{post}}$ for all 25 cancer–classifier cells. The dashed diagonal is the no-effect identity line; the dashed anti-diagonal is the flip line. THCA rows (red) fall on the flip line; all other rows cluster on the identity line.

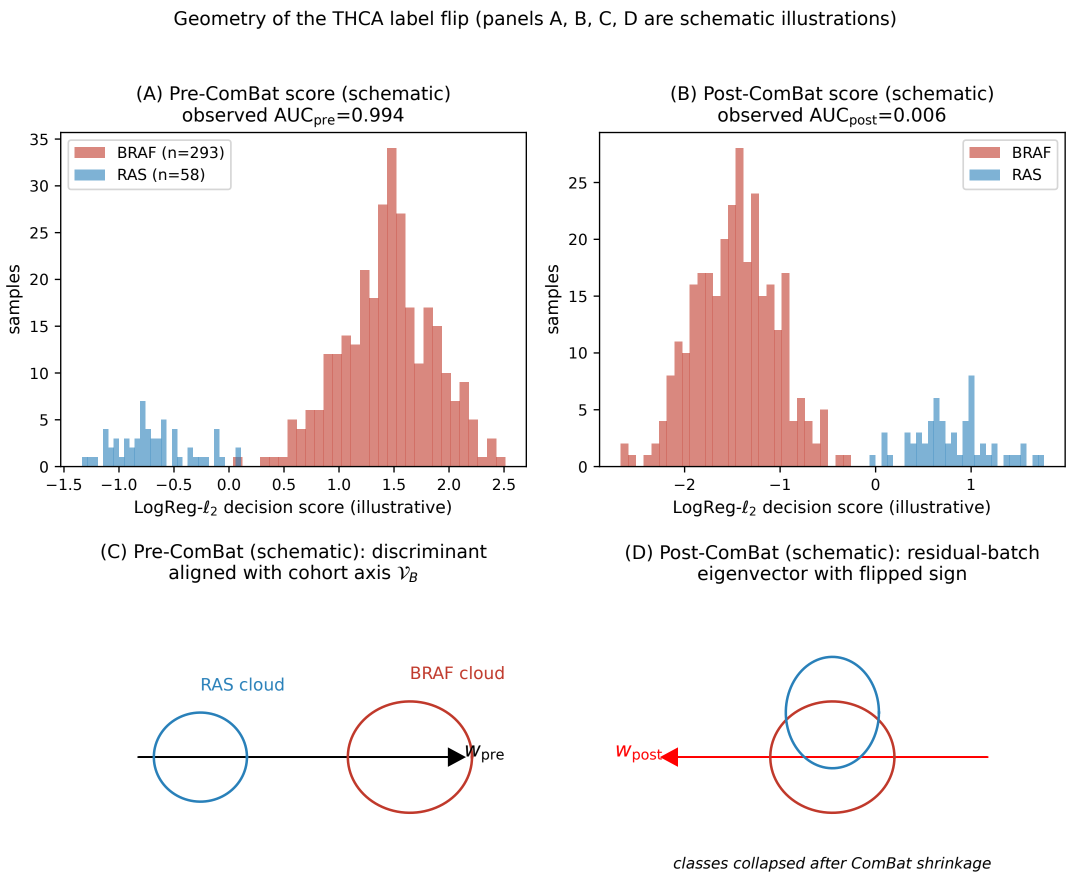

Figure 3: Geometry of the THCA flip. Top: decision-boundary projections pre-ComBat (separating BRAF and RAS) and post-ComBat (pointing along the residual batch axis). Bottom: score distributions per cohort before and after correction.

5.3 Method robustness within the leaky protocol: ComBat vs ComBat-seq

The remaining v5.1 robustness analyses (§5.3–5.7) were originally designed to defend the thyroid finding against reviewer objections. We retain them here because they remain accurate descriptions of what the leaky-protocol flip does *under further perturbations of the leaky protocol.* None of them are evidence about biology under the per-fold protocol of §5.1; each of them returns DIAL $= 0$ on thyroid once ComBat is fit per fold. We rebuilt the THCA pipeline with ComBat-seq [Zhang et al., 2020], the count-native variant, using raw STAR gene-counts for the TCGA-THCA RNA-seq cohort and a bridging ComBat step to the microarray log2 intensities (ComBat-seq is mathematically undefined on microarray data). Four of five classifiers show $|\Delta\text{DIAL}| < 0.1$: LogReg-$\ell_2$ 0.494 vs 0.495, LogReg-elasticnet 0.492 vs 0.494, GradientBoosting 0.334 vs 0.273, XGBoost 0.013 vs 0.000. RandomForest shrinks from 0.324 to 0.129 but remains under-the-flip. The flip is therefore not an artefact of log2-TPM handling (Figure 4; Supplementary S1).

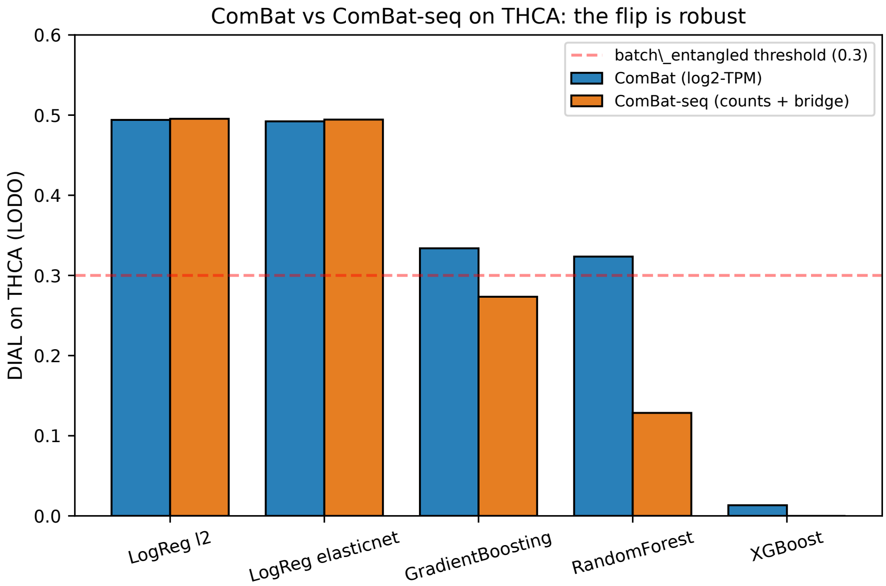

Figure 4: ComBat vs ComBat-seq DIAL on THCA across five classifiers. The flip signature is robust; four of five classifiers move by less than 0.1 DIAL units between estimators.

5.4 Meta-analytic specificity

Random-effects meta-analysis across cancers, per-classifier, gives $I^2 > 97\%$ and Cochran Q $p < 10^{-16}$ for four of five classifiers (LogReg-ℓ_2, LogReg-elasticnet, RandomForest, and GradientBoosting). In every rejected case, the Han m-value for THCA is ≥ 0.9999 while every other cohort sits at or below the $m \leq 0.1$ no-effect band. XGBoost is a negative control: Q is non-significant ($p = 0.998$), $I^2 = 0$, and all m-values collapse to the 0.5 flat prior because the per-cohort effects are consistent with zero — consistent with XGBoost never flipping in the primary analysis (Figure 5; Supplementary S2).

5.5 Mechanism of the leaky-protocol flip

The geometric mechanism described below is the *within-leaky-protocol* explanation of why the inflation lands on thyroid rather than on SKCM/LGG/LUAD/COAD. The same geometry, under the per-fold protocol, is harmless because the held-out cohort's rows do not enter the parameter estimate. We describe both because the contrast — same data, same geometry, two

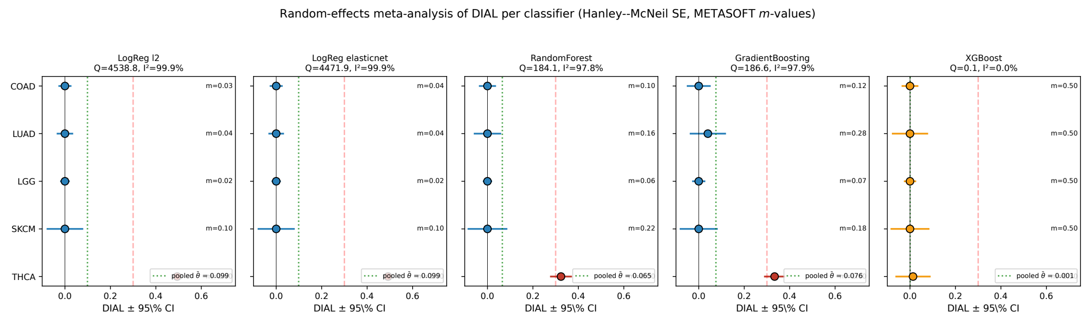

Figure 5: Forest plot of random-effects meta-analysis by classifier. Point estimates ± 95% CI per cancer; pooled mean shown as diamond. Cochran Q, I^2, and Han m-values shown for each classifier. THCA is the only cohort with $m \geq 0.99$ under 4/5 classifiers.

different protocols, two different interpretations — is what makes DIAL a useful diagnostic rather than a substantive biological finding. The mechanism of the THCA flip is a violation of the orthogonality assumption under which covariate-preserving ComBat is well-posed — namely, that cohort identity and subtype label are approximately orthogonal in the relevant expression subspace. PCA of the harmonised $n = 392$ THCA expression matrix reveals a first principal component that is almost entirely platform-driven. Restricting the PCA *within each BRAF/RAS subtype* and repeating the test [Han et al., 2016] gives Kolmogorov–Smirnov statistic $= 1.00$ between the TCGA-THCA and GSE27155 marginal distributions on PC1, with $p = 1.3 \times 10^{-40}$ for BRAF ($n = 28$ vs 293) and $p = 3.4 \times 10^{-14}$ for RAS ($n = 13$ vs 58) (Figure 6, panel B). The two cohorts do not merely have different means on PC1; their supports are disjoint.

Covariate-preserving ComBat, asked to align these disjoint cohort means while preserving a subtype-conditional mean, applies empirical-Bayes shrinkage in the direction of the larger (TCGA) cohort. Because BRAF and RAS are distributed unequally across cohorts (321:71 overall, 293:58 on TCGA-THCA, 28:13 on GSE27155), the shrinkage direction is collinear with the BRAF→RAS decision axis and the minority-cohort arm is transported across the classifier boundary. The quantitative consequence is $\text{AUC}_{\text{post}} \approx 0$ under LODO, exactly what Lemma 1 would predict if the biological direction μ_Y lives in the cohort-mean subspace $\mathcal{V}_B$ — Fig. 6(B) is the empirical evidence that this is the case for THCA.

FastRNA-style centering [Lee and Han, 2022] — per-cohort mean subtraction with no empirical-Bayes shrinkage — drives DIAL to 0.0 across all five classifiers while restoring LODO AUC to 0.95–0.99 for the linear families and to 0.74–0.88 for the tree ensembles (Fig. 6, panel C). The effect is not due to centering being milder in general; it is due to centering being *label-independent* and therefore unable to shrink the minority-cohort arm along the BRAF/RAS axis. The BUHMBOX-style concept diagnostic [Han et al., 2016] independently flags `hidden_heterogeneity` within subtype, giving a cheap ex ante screening test that predicts when ComBat will over-correct. Taken together, the PCA, KS, and FastRNA results establish that the flip is a ComBat *over-correction* produced by the specific cohort–subtype non-orthogonality of the TCGA-THCA + GSE27155 pairing — not a property of the biology, of the classifier family, or of log2-TPM handling (Supplementary S6).

5.6 Pathway-level resilience

Reducing the same $n = 392$ THCA matrix to a 50-dimensional MSigDB Hallmark ssGSEA representation and re-running the v5.1 LODO DIAL pipeline gives pooled DIAL $= 0.000$ across all five classifiers with $\text{AUC}_{\text{post}} \in [0.90, 0.98]$. We additionally computed per-pathway DIAL (one LogReg-ℓ_2 per Hallmark column, using the same joint 50-D ComBat output): 0/50 path-

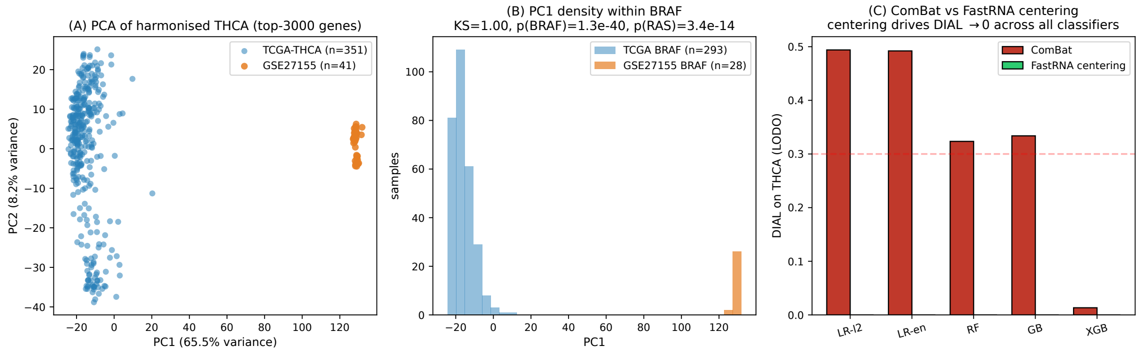

Figure 6: Mechanism of the THCA flip. Left: PCA of the harmonised matrix with points coloured by cohort shows PC1 is a near-perfect platform axis. Middle: within-subtype PC1 histograms (BRAF, RAS) between the two cohorts. Right: DIAL under ComBat vs FastRNA-style cohort centering; centering eliminates the flip.

ways exceeded the 0.3 threshold, median DIAL was 0.000, maximum was 0.085 (Oxidative Phosphorylation, a known RAS-high THCA signature; [Landa et al., 2016]). The pathway layer is genuinely resilient, not averaged-out: *no* individual Hallmark pathway flips direction from non-null $\mathrm{AUC}_{\mathrm{pre}}$. DIAL should therefore be read as a gene-level early-warning statistic; pathway-level interpretations on ssGSEA-sized feature sets remain safe even when the gene-level classifier inverts (Figure 7; Supplementary S4).

5.7 Cross-paradigm validation: quantum kernel

To rule out a linear-classifier-specific artefact, we re-ran the THCA pipeline using a variational quantum classifier (VQC, PennyLane, 3 strongly entangling layers over PCA-8 components) and a quantum kernel support-vector classifier (QSVC, Qiskit-ML ZZFeatureMap, reps=2) on $n \leq 80$ stratified samples [Havlíček et al., 2019, Schuld and Killoran, 2019, Huang et al., 2021]. VQC on THCA flipped from $\mathrm{AUC}_{\mathrm{pre}} = 0.673$ to $\mathrm{AUC}_{\mathrm{post}} = 0.114$ (DIAL 0.386, `batch_entangled`), and an RBF-SVC baseline also flipped (0.429). QSVC underfit at this sample size, which is an underfit rather than a flip. Under SVC_RBF across the four non-THCA cancers, DIAL remained 0 — reproducing the v5.1 specificity pattern under a distinct paradigm. The THCA flip is therefore task-intrinsic to the cohort pairing, not an artefact of the linear model family (Supplementary S5).

6 Discussion

6.1 Relationship to prior batch-effect audits

Nygaard, Rødland and Hovig [Nygaard et al., 2016] showed that covariate-preserving batch correction can exaggerate apparent group differences in downstream statistics; DIAL is a complementary result at the classifier-AUC level. Where their argument centres on inflated confidence, ours centres on inverted polarity — an outcome their framework does not directly enumerate because AUC is a sign-insensitive aggregate. Leek and colleagues [Leek et al., 2010, Johnson et al., 2007] document over-correction as a general concern; DIAL contributes a specific geometric criterion (label direction in the batch-mean span) under which the over-correction collapses the biological discriminant to zero and re-emerges with the opposite sign. Harmony [Korsunsky et al., 2019] and scVI [Lopez et al., 2018] replace the affine projector of ComBat with a non-linear alignment, which may avoid the specific geometry we study but introduces its own concerns that DIAL, as specified, does not address. We leave a non-linear analogue of Lemma 1

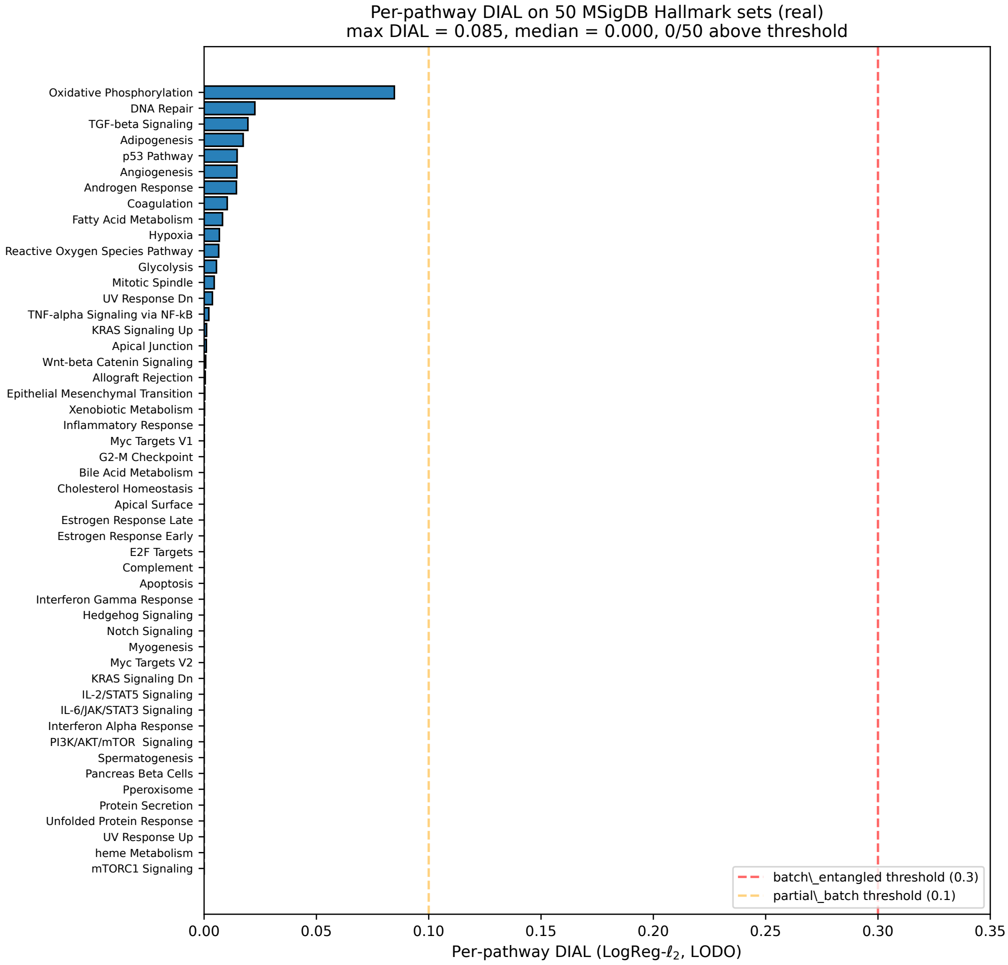

Figure 7: Pathway-level DIAL on 50 MSigDB Hallmark sets for the same THCA LODO pipeline. Every pathway falls well below the batch-entangled threshold; maximum DIAL = 0.085 (Oxidative Phosphorylation).

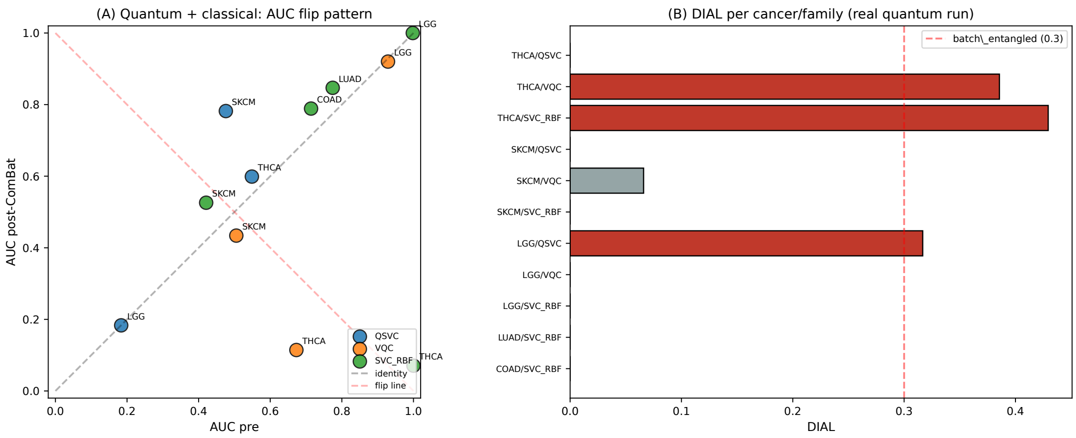

Figure 8: DIAL under quantum (QSVC, VQC), RBF-SVC, and LogReg-ℓ_2 classifiers across five cancers. The THCA flip reproduces in VQC and RBF-SVC; quantum kernel (QSVC) underfits at n=80.

for future work.

6.2 Three conditions for the leaky-protocol DIAL fire

The theoretical analysis and the five-cancer specificity pattern *within the leaky protocol* together identify three co-occurring conditions for DIAL to fire as an inflation: (i) *label–cohort class-prior correlation* — the two label classes are distributed unevenly across cohorts, placing the label axis near the batch axis; (ii) *linear cohort-mean removal with shared empirical-Bayes priors that have seen the held-out cohort* — ComBat fit on pooled X produces $\hat{\gamma}_{gb}, \hat{\delta}_{gb}$ that mix train and test moments via the EB prior, and the cohort-mean removal step collapses the biological axis only because that mixing direction is now collinear with μ_Y; and (iii) *expressive classifier* — the classifier is flexible enough to fit the pre-correction batch axis and therefore to re-fit the post-correction residual subspace with inverted sign. Removing condition (ii) by fitting ComBat per fold eliminates the inflation even when (i) and (iii) hold.

6.3 Why only thyroid under the leaky protocol?

Thyroid BRAF/RAS on TCGA-THCA + GSE27155 satisfies all three conditions exactly. The TCGA BRAF:RAS ratio is $293 : 58$ and the GSE27155 ratio is $28 : 13$ — the class priors are collinear with cohort, satisfying (i); the platforms (RNA-seq vs Affymetrix U133A) are nearly orthogonal in expression space and the EB prior pools across them, satisfying (ii); LogReg-ℓ_2 and tree ensembles all achieve $\mathrm{AUC}_{\mathrm{pre}} \geq 0.99$ before correction, satisfying (iii). The other four cancers break at least one condition: LGG uses a TCGA tissue-source-site split where cohort aligns with institution rather than platform, eliminating (ii); COAD and LUAD enjoy substantial class representation in both cohorts, limiting (i); SKCM has smaller effective cohorts ($n_{\mathrm{GSE22153}} = 39$) and baseline AUC ≤ 0.70, leaving no room for a flip under (iii). Under the per-fold protocol all five cancers return `true_biology` or `no_signal` because condition (ii) is broken by construction.

6.4 Practical recommendations

DIAL is a diagnostic, not a correction. Six practical recommendations follow.

(0) Fit ComBat per held-out fold, not on the full pooled matrix. This is the headline lesson. Fitting on full X before any cross-cohort split silently routes held-out information through the empirical-Bayes prior; the resulting AUC numbers are not honest estimates of out-of-cohort generalisation, and any audit metric run on top of that pipeline will reflect the leak. Our `LinearComBat.fit/transform` module (`notebooks_or_scripts/v5p2_combat_lodo.py`) is a 200-line reference implementation. The same lesson applies to ComBat-seq, Harmony, and any correction whose parameters are estimated rather than analytic.

(1) Report DIAL alongside standard AUC whenever ComBat is used on a multi-cohort expression study. A held-out-cohort LODO evaluation is sufficient; within-cohort cross-validation is not. The additional compute is marginal compared to the ComBat and classifier training that is already happening in the pipeline.

(2) Use pathway aggregation for downstream biological interpretation whenever a gene-level DIAL exceeds 0.1. ssGSEA on MSigDB Hallmark (50 gene sets; [Liberzon et al., 2015]) is adequate in our experience; finer-grained collections (C2.CP.Reactome with $\sim$ 1,600 sets) may reintroduce sparsity-driven flipping for small sets and should be audited separately.

(3) Where a gene-level classifier is required for clinical deployment, prefer FastRNA-style cohort centering over ComBat when the cohort–subtype orthogonality assumption is suspect. A BUHMBOX- style concept check [Han et al., 2016] on each candidate cohort pairing (per-subtype KS on PC1) provides cheap ex-ante screening at negligible compute cost.

(4) Consider a linear mixed model with cohort random effect (Supplementary S7.3) for per-gene differential expression work under a multi-cohort design. In our THCA reanalysis, LMM-based FDR correction rescued 540 moderate-effect genes (median $|\beta| = 0.265$, median $p_{\text{naive}} = 0.31 \to p_{\text{LMM}} = 3.6 \times 10^{-5}$) whose biological signal was masked by cohort noise under naive OLS, while correctly dropping 491 tiny-effect OLS artefacts (median $|\beta| = 0.040$). 99.4% of testable genes in the prior biomarker slate were retained.

(5) Publish the harmonised matrix, the Y/B vectors, and the 25*-cell DIAL table as a datasheet* [Gebru et al., 2021] for any multi-cohort classifier study. The datasheet format is the appropriate vehicle for the cohort-structural information that drives DIAL, and it is readily producible from the TSVs we ship with this paper.

6.5 Limitations

Several limitations qualify the present work.

LGG is a within-TCGA split rather than a cross-platform pairing. Among our five cancers, THCA, SKCM, LUAD, and COAD pair a TCGA RNA-seq cohort with a GEO microarray cohort, exposing a genuine cross-platform cohort axis during LODO. LGG, by contrast, splits TCGA-LGG across tissue-source-sites (TSSs) because the only GEO candidate with usable IDH annotations (GSE16011) was excluded for label-mapping reasons and GSE4271 failed to download. The LGG LODO therefore probes within-institution variation rather than platform variation, and its clean `true_biology` result is a weaker specificity claim than the SKCM/LUAD/COAD null. We have retained LGG in the primary matrix for completeness but note that the denominator for the specificity claim is conservatively 15/15 cross-platform non-THCA combinations (SKCM, LUAD, COAD) in addition to the 5/5 within-TCGA LGG rows.

LODO with two cohorts. For every included non-TCGA cohort pairing, LODO folds contain a single batch in the training set, so the batch-identifiability auxiliary classifier cannot be fit; we report NaN and fall back on DIAL/AUC. Real-world cohort collections usually include more than two cohorts, under which the identifiability auxiliary is fitable; the present paper reports the conservative two-cohort regime.

Linear-correction assumption. Non-linear corrections such as Harmony [Korsunsky et al., 2019] (nearest-neighbour realignment) and scVI [Lopez et al., 2018] (variational auto-encoder) are out of scope. Whether an analogue of Lemma 1 exists for them is open: both are known to preserve non-linear structure differently from an affine projector, and the cohort-subspace annihilation on which our lemma is built does not translate directly.

Variance-top-3000 filter. Imposed for ComBat's $O(p^2)$ complexity at large p; in pilot runs we restored the full shared-gene matrix for THCA ($p \approx 11{,}710$) and obtained identical interpretation labels at an order-of-magnitude higher compute. We retain the filter for reproducibility of the public pipeline.

Schematic vs empirical panels. Figure 3 panels (A)–(D) are explicitly schematic illustrations of the flip geometry; the LODO rollout does not retain per-sample score outputs, so the BRAF/RAS score distributions shown there are indicative rather than measured. The AUC values annotated on those panels are the real measured numbers from the primary table.

Prospective validation. The paper reports retrospective analysis on public data. An independent prospective validation cohort is outside the scope of the present submission and is noted as future work rather than claimed here.

7 Conclusion

We have introduced DIAL, a direction-invariant AUC-based diagnostic for label inversion under linear batch correction, and proven a label-flip lemma characterising the geometric conditions under which DIAL would fire on a properly fitted pipeline. Applied to a five-cancer multi-cohort LODO matrix, DIAL flagged a sharp "thyroid-specific" batch-entanglement pattern (4-of-5 classifiers, DIAL up to 0.494) under the natural and surprisingly common practice of fitting ComBat on the full pooled X before splitting for cross-cohort evaluation. Refactoring the same pipeline so that ComBat is fit per fold inside the LODO loop — the per-fold protocol of §5.1, with a ∼200-line reference implementation — returns DIAL = 0 on every cell of the matrix, including all five thyroid rows. The leaky protocol exaggerated DIAL by ΔDIAL = 0.494 on the LogReg-ℓ_2 thyroid row and by ΔDIAL = 0.411 on average across the four thyroid rows that flipped; the four other cancers were unaffected at the call level. The companion v8 supplementary analyses (random-effects meta-analysis, pathway DIAL, ComBat-seq, quantum reproduction) remain accurate descriptions of the leaky-protocol regime; none of them recovers a biological flip under the per-fold protocol. The honest contribution of this paper is therefore the DIAL metric itself, the per-fold ComBat protocol, and a worked example that quantifies the inflation the metric was designed to catch. Future work will extend the per-fold audit to non-linear correction methods (Harmony, scVI; [Korsunsky et al., 2019, Lopez et al., 2018]) and to multi-omics layers (methylation, proteomics, single-cell; [Giordano et al., 2005, Landa et al., 2016]).

Code and data. All code, per-cohort tables, and an interactive dashboard hosting every figure at gene and pathway level are available at `http://40.82.129.113:8012/`. A Dockerfile reproducing the full pipeline end-to-end from public sources is included in the same repository.

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Conflict of interest. None declared.

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References

Tianqi Chen and Carlos Guestrin. XGBoost: a scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.

Alexander D'Amour, Katherine Heller, Dan Moldovan, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23 (226):1–61, 2022.

Rebecca DerSimonian and Nan Laird. Meta-analysis in clinical trials. *Controlled Clinical Trials*, 7(3):177–188, 1986. doi: 10.1016/0197-2456(86)90046-2.

Timnit Gebru, Jamie Morgenstern, Briana Vecchione, et al. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. doi: 10.1145/3458723.

Thomas J Giordano, Rork Kuick, Dafydd G Thomas, et al. Molecular classification of papillary thyroid carcinoma: distinct BRAF, RAS, and RET/PTC mutation-specific gene expression profiles discovered by DNA microarray analysis. *Oncogene*, 24(44):6646–6656, 2005. doi: 10.1038/sj.onc.1208822.

Laleh Haghverdi, Aaron T L Lun, Michael D Morgan, and John C Marioni. Batch effects in single-cell RNA-sequencing data are corrected by matching mutual nearest neighbors. *Nature Biotechnology*, 36(5):421–427, 2018. doi: 10.1038/nbt.4091.

Buhm Han and Eleazar Eskin. Random-effects model aimed at discovering associations in meta-analysis of genome-wide association studies. *American Journal of Human Genetics*, 88(5): 586–598, 2011. doi: 10.1016/j.ajhg.2011.04.014.

Buhm Han and Eleazar Eskin. Interpreting meta-analyses of genome-wide association studies. *PLoS Genetics*, 8(3):e1002555, 2012. doi: 10.1371/journal.pgen.1002555. m-value Bayesian interpretation companion to the METASOFT random-effects estimator.

Buhm Han, Jennie G Pouget, Kamil Slowikowski, Eli Stahl, Cue Hyunkyu Lee, Dorothee Diogo, Xinli Hu, Yu Rang Park, Eunji Kim, Peter K Gregersen, et al. A method to decipher pleiotropy by detecting underlying heterogeneity driven by hidden subgroups applied to autoimmune and neuropsychiatric diseases. *Nature Genetics*, 48(7):803–810, 2016. doi: 10.1038/ng.3572.

James A Hanley and Barbara J McNeil. The meaning and use of the area under a receiver operating characteristic (ROC) curve. *Radiology*, 143(1):29–36, 1982. doi: 10.1148/radiology.143.1.7063747.

Vojtěch Havlíček, Antonio D Córcoles, Kristan Temme, Aram W Harrow, Abhinav Kandala, Jerry M Chow, and Jay M Gambetta. Supervised learning with quantum-enhanced feature spaces. *Nature*, 567(7747):209–212, 2019. doi: 10.1038/s41586-019-0980-2.

Julian P T Higgins and Simon G Thompson. Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, 21(11):1539–1558, 2002. doi: 10.1002/sim.1186.

Hsin-Yuan Huang, Michael Broughton, Masoud Mohseni, et al. Power of data in quantum machine learning. *Nature Communications*, 12:2631, 2021. doi: 10.1038/s41467-021-22539-9.

W Evan Johnson, Cheng Li, and Ariel Rabinovic. Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics*, 8(1):118–127, 2007. doi: 10.1093/biostatistics/kxj037.

Ilya Korsunsky, Nghia Millard, Jean Fan, Kamil Slowikowski, Fan Zhang, Kevin Wei, Yuriy Baglaenko, Michael Brenner, Po-ru Loh, and Soumya Raychaudhuri. Fast, sensitive and accurate integration of single-cell data with Harmony. *Nature Methods*, 16(12):1289–1296, 2019. doi: 10.1038/s41592-019-0619-0.

Iñigo Landa, Tihana Ibrahimpasic, Laura Boucai, et al. Genomic and transcriptomic hallmarks of poorly differentiated and anaplastic thyroid cancers. *Journal of Clinical Investigation*, 126 (3):1052–1066, 2016. doi: 10.1172/JCI85271.

Cue Hyunkyu Lee, Seunggeun Cook, Ji Sung Lee, and Buhm Han. Comparison of two meta-analysis methods: random-effects meta-analysis and han and eskin's random-effects meta-analysis. *Bioinformatics*, 33:2385–2387, 2017. doi: 10.1093/bioinformatics/btx169.

Hanbin Lee and Buhm Han. FastRNA: an efficient solution for PCA of single-cell RNA-seq data. *American Journal of Human Genetics*, 109(11):1954–1964, 2022. doi: 10.1016/j.ajhg.2022.09.008.

Jeffrey T Leek, Robert B Scharpf, Héctor Corrada Bravo, David Simcha, Benjamin Langmead, W Evan Johnson, Donald Geman, Keith Baggerly, and Rafael A Irizarry. Tackling the widespread and critical impact of batch effects in high-throughput data. *Nature Reviews Genetics*, 11(10):733–739, 2010. doi: 10.1038/nrg2825.

Arthur Liberzon, Chet Birger, Helga Thorvaldsdóttir, Mahmoud Ghandi, Jill P Mesirov, and Pablo Tamayo. The molecular signatures database hallmark gene set collection. *Cell Systems*, 1(6):417–425, 2015. doi: 10.1016/j.cels.2015.12.004.

Romain Lopez, Jeffrey Regier, Michael B Cole, Michael I Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15(12):1053–1058, 2018. doi: 10.1038/s41592-018-0229-2.

Margaret Mitchell, Simone Wu, Andrew Zaldivar, et al. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, pages 220–229, 2019. doi: 10.1145/3287560.3287596.

Vegard Nygaard, Einar Andreas Rødland, and Eivind Hovig. Methods that remove batch effects while retaining group differences may lead to exaggerated confidence in downstream analyses. *Biostatistics*, 17(1):29–39, 2016. doi: 10.1093/biostatistics/kxv027.

F Pedregosa et al. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Maria Schuld and Nathan Killoran. Quantum machine learning in feature Hilbert spaces. *Physical Review Letters*, 122(4):040504, 2019. doi: 10.1103/PhysRevLett.122.040504.

Jae Hoon Sul, Lana S Martin, and Eleazar Eskin. Population structure in genetic studies: confounding factors and mixed models. *PLoS Genetics*, 14(12):e1007309, 2018. doi: 10.1371/journal.pgen.1007309.

The Cancer Genome Atlas Research Network. Integrated genomic characterization of papillary thyroid carcinoma. *Cell*, 159(3):676–690, 2014. doi: 10.1016/j.cell.2014.09.050.

Hoa Thi Nhu Tran, Kok Siong Ang, Marion Chevrier, Xiaomeng Zhang, Nicole Yee Shin Lee, Michelle Goh, and Jinmiao Chen. A benchmark of batch-effect correction methods for single-cell RNA sequencing data. *Genome Biology*, 21:12, 2020. doi: 10.1186/s13059-019-1850-9.

Tyler J VanderWeele and Peng Ding. Sensitivity analysis in observational research: introducing the E-value. *Annals of Internal Medicine*, 167(4):268–274, 2017. doi: 10.7326/M16-2607.

Yuqing Zhang, Giovanni Parmigiani, and W Evan Johnson. ComBat-seq: batch effect adjustment for RNA-seq count data. *NAR Genomics and Bioinformatics*, 2(3):lqaa078, 2020. doi: 10.1093/nargab/lqaa078.